

|    |   |
|----|---|
| 13 | B |
|----|---|

|    |   |
|----|---|
| 14 | C |
|----|---|

|    |   |
|----|---|
| 17 | B |
|----|---|

|    |   |
|----|---|
| 19 | A |
|----|---|

|    |   |
|----|---|
| 20 | C |
|----|---|

|    |   |
|----|---|
| 21 | B |
|----|---|

|    |                                                                                                                                                                                                                                                                                  |   |                                                                                                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------|
|    | only transverse waves can be polarised ✓                                                                                                                                                                                                                                         |   |                                                                                                                      |
| 02 | <p>Two from: ✓ ✓<br/> use of <math>W=Pt</math><br/> use of <math>W=QV</math><br/> converting t to seconds</p> <p>1100 (C) ✓</p> <p><b>OR</b></p> <p>Two from: ✓ ✓<br/> use of <math>P=IV</math><br/> use of <math>Q=It</math><br/> converting t to seconds</p> <p>1100 (C) ✓</p> | 3 | <p>Condone time in minutes for 1<sup>st</sup> MP only.</p> <p>Condone time in minutes in 2<sup>nd</sup> MP only.</p> |

|    |                                                                                                                                                                                                                                                                                             |   |                                          |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------|
|    | $5.64 \times 10^{-10} \text{ (m)} \checkmark$                                                                                                                                                                                                                                               |   | Accept $5.6 \times 10^{-10} \text{ (m)}$ |
| 02 | $d$ calculated (= 4 $\mu\text{m}$ ) <b>or</b> appropriate use of $\frac{1}{250 \times 10^3} \checkmark$<br>use of $n\lambda = d \sin \theta$<br>(= $6.5 \times 10^{-7} \text{ m}$ ) $\checkmark$<br>use of $c = f\lambda \checkmark$<br>$f = 4.6(4) \times 10^{14} \text{ (Hz)} \checkmark$ | 4 | Condone power of 10 for MP1. MP2 and MP3 |



| Question | Marking guidance                                                                                                                        | Additional comments/guidelines                                                                                                                                                                                         | Mark |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 02.1     | The maximum angle of <u>incidence</u><br>at which refraction occurs ✓                                                                   | Accept “incidence” explained.<br>Condone minimum angle of incidence at which total internal reflection occurs<br>Accept the angle of incidence that produces an angle of refraction of $90^\circ$ / along the boundary | 1    |
| 02.2     | Use of $\sin\theta = \frac{n_2}{n_1}$ to find $n_2$ (= 1.3) ✓<br>Use of $n = \frac{c}{v}$ ✓<br>$2.3 \times 10^8$ (m s <sup>-1</sup> ) ✓ | For max 1 mark, condone $1.52 = \frac{3.0 \times 10^8}{v}$                                                                                                                                                             | 3    |

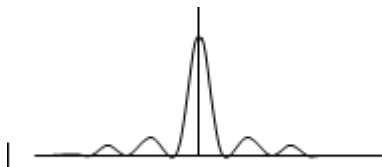
|             |                                                                                                                                          |                                                                             |   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---|
|             | interference (between light from different slits) ✓<br>Idea that different wavelengths have maxima at different angles ✓                 | maxima / minima                                                             |   |
| <b>04.2</b> | correct read-off of $\lambda$ from the diagram = 619 (nm) ✓<br>use of $E_{\gamma} = \frac{hc}{\lambda}$ ✓<br>$3.2 \times 10^{-19}$ (J) ✓ | Ignore POT error for mp1<br><br>Allow 2 marks max if 618 or 620nm are used. | 3 |

| Question | Marking guidance | Additional comments/guidelines | Mark |
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| Question | Marking guidance                                                                                                                       | Additional comments/guidelines                                                                                   | Mark |
|----------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|
| 06.1     | Use of $W = QV$ ✓<br>9400 V ✓                                                                                                          |                                                                                                                  | 2    |
| 06.2     | Use of $E_k = \frac{1}{2}mv^2$ ✓<br>Use of $p = mv$ ✓<br>$5.2 \times 10^{-23}$ (kg m s <sup>-1</sup> ) to at least 2 sf ✓              |                                                                                                                  | 3    |
| 06.3     | use of $\lambda = h/p$ ✓<br>$n = 2$ seen or implied ✓<br>use of (n) $\lambda = d \sin \theta$ ✓<br>$d = 1.5 \times 10^{-10}$ (m) cao ✓ | condone incorrect $n$ for MP3<br>If $5 \times 10^{-23}$ kg m s <sup>-1</sup> used, $d = 1.5 \times 10^{-10}$ (m) | 4    |

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| Question | Marking guidance | Additional comments/guidelines | Mark |
|----------|------------------|--------------------------------|------|
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| Question | Answers                                                                                                                                                                                                                                                                                                                                           | Additional comments/Guidelines                                                                                                                                                                                    | Mark | ID details |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|
| 03.1     | <p>Correct general shape with correct curvature of lines, minima at zero or constant value above and symmetrical</p> <p>AND</p> <p>each maximum lower than the previous one ✓</p> <p>Width of central maximum approximately <math>2 \times</math> width of subsequent maxima and central maximum much higher than the 1st max on both sides ✓</p> | <p>Must be 5 maxima for MP2</p>  <p>For MP2 look for central maximum at least 3 times as high as the 1st max on both sides</p> | 2    | E          |

| Question | Answers                                                                              | Additional comments/Guidelines | Mark | ID details |
|----------|--------------------------------------------------------------------------------------|--------------------------------|------|------------|
| 03.2     | M marked on the abscissa between the 1st maximum from Question 03.1 and the origin ✓ |                                | 1    | E          |

| Question     | Answers           | Additional comments/Guidelines                                                                                          | Mark     | ID details |
|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------|----------|------------|
| 03.3         | Brighter maxima ✓ | <p>Allow increase (intensity)</p> <p>Condone 'stronger'</p> <p>Treat comments about position and clarity as neutral</p> | 1        | E          |
| <b>Total</b> |                   |                                                                                                                         | <b>4</b> |            |

| Question | Answers | Additional comments/Guidelines | Mark | ID<br>details |
|----------|---------|--------------------------------|------|---------------|
|----------|---------|--------------------------------|------|---------------|

| Question | Answers                                                                                      | Additional comments/Guidelines                                                                                | Mark | ID details |
|----------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------|------------|
| 05.1     | Multiplies by $1.6 \times 10^{-19}$ to give $5.47(2) \times 10^{-19}$ (J) to at least 3 sf ✓ | must show some working<br>do not allow $5.48 \times 10^{-19}$ (J)<br>do not allow route with non-zero current | 1    |            |

| Question | Answers                                                                                                              | Additional comments/Guidelines                                                                              | Mark | ID details |
|----------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------|------------|
| 05.2     | Uses $hf = \phi + E_{k(\max)}$ ✓<br>Uses $c = f\lambda$ or $E = \frac{hc}{\lambda}$ ✓<br>$2.15 \times 10^{-7}$ (m) ✓ | Need consistent units in MP1<br>Look for $hf = 9.248 \times 10^{-19}$ J or 5.56 eV<br>Accept 2.2 and 2.1(4) | 3    | E          |

| Question | Answers                                                                                                                                         | Additional comments/Guidelines                                                                                                                                                                          | Mark | ID details |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|
| 05.3     | Gradient for both (new) lines the same as that for sodium ✓<br><br>Magnesium line lower than sodium AND Caesium line higher than sodium line. ✓ | Condone lack of labels in MP1<br>caesium line must have a negative intercept on ordinate<br>Condone Ca for Caesium<br>Ignore any lines drawn below horizontal axis<br>Do not award MP2 if MP1 incorrect | 2    | E          |

| Question     | Answers                                                                                                                                                           | Additional comments/Guidelines                                                                                                                                                                                                                         | Mark     | ID details |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|
| 05.4         | <p><math>V_s</math> reduced because</p> <p>(photo)electrons collide with atoms in the air✓</p> <p>Reducing their KE / their velocity/ <math>KE_{\max}</math>✓</p> | <p>accept argument that <math>V</math> becomes zero because <u>all</u> photoelectrons are stopped</p> <p>Allow idea that KE is absorbed or used up</p> <p>Do not condone 'energy'</p> <p>Do not give MP2 if process is confused with work function</p> | 2        | E          |
| <b>Total</b> |                                                                                                                                                                   |                                                                                                                                                                                                                                                        | <b>8</b> |            |



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| Question | Answers | Additional comments/Guidelines | Mark | ID details |
|----------|---------|--------------------------------|------|------------|
|----------|---------|--------------------------------|------|------------|

| Question     | Answers                                                                                                                                                                                                                                           | Additional comments/Guidelines                                                                                   | Mark     | AO                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------|--------------------|
| <b>02</b>    | Use of $d \sin \theta = n\lambda$ ✓<br>$d = 3.5 \times 10^{-6} \text{ (m)}$ <b>OR</b> $\frac{42 \times 10^{-3}}{12000}$ ✓<br>uses $n = 2$ <b>OR</b> $\theta = 20.7^\circ$ ✓<br>$6.2 \times 10^{-7} \text{ (m)}$ <b>OR</b> $6.19 \times 10^{-7}$ ✓ | Condone pot error in MP2<br><br>Answer of $1.16 \times 10^{-6} \text{ m}$ from $\theta = 41.4^\circ$ gets max 3. | 4        | 2 × AO1<br>2 × AO2 |
| <b>Total</b> |                                                                                                                                                                                                                                                   |                                                                                                                  | <b>4</b> |                    |

| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question     | Answers                                                                                                                                                                                                                                                                                          | Additional comments/Guidelines                                              | Mark     | AO                         |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|----------------------------|
| <b>03</b>    | <p>Max 3<br/>Lamps too far apart (to produce fringes that are observable to the naked eye) ✓</p> <p>Not coherent... ✓<br/>...because they don't have a constant phase relationship ✓</p> <p>Idea that a pair of wavefronts or photons coinciding are not likely to have the same frequency ✓</p> | For MP4, accept the idea that filament lamps emit light of many frequencies | 3        | <p>1 × AO1<br/>2 × AO2</p> |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                  |                                                                             | <b>3</b> |                            |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mark | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02       | <p><math>V</math> is (terminal) potential difference AND <math>\epsilon</math> is emf (of the battery) ✓</p> <p>Any statement that voltage is the work done/energy transferred per unit charge wtte ✓</p> <p><math>\epsilon</math> is the work done/energy per unit charge moving charge through the complete circuit(including the internal resistance)/provided by source</p> <p>OR</p> <p><math>V</math> is work done/energy per unit charge moving charge through <math>R</math> / the external circuit✓</p> | 3    | <p>Condone 'electromotive force' for emf and p.d. for potential difference/drop</p> <p>BOD for poor QWC but <b>not</b> if it is ambiguous (e.g. electric motor force)</p> <p>"energy per unit charge" is insufficient for MP2</p> <p>Condone use of 'through' for 'across'.</p> <p>Accept 'available to <math>R</math>'</p> <p>Treat incorrect but not-contradictory comments about <math>\epsilon</math> or <math>V</math> as neutral.</p> |
| Total    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3    |                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Question | Marking guidance | Mark | Comments |
|----------|------------------|------|----------|
|----------|------------------|------|----------|

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|----|----------|---------------------------|
| 20 | <b>D</b> | polarisation at the slits |
|----|----------|---------------------------|

|    |          |       |
|----|----------|-------|
| 21 | <b>C</b> | 14.5° |
|----|----------|-------|



| Question | Answers                                                                                | Additional comments/Guidelines                                                                          | Mark | AO                 |
|----------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|--------------------|
| 03       | Use of $\sin C = 1/n$ ✓<br><br>Diamond has (much) larger refractive index than glass ✓ | If values of n are determined at least one must be correct to award MP1.<br><br>Accept reverse argument | 2    | 1 × AO2<br>1 × AO3 |
| Total    |                                                                                        |                                                                                                         | 2    |                    |

| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers                                                                                        | Additional comments/Guidelines                                                                                      | Mark | AO                 |
|----------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------|--------------------|
| 09.1     | Use of tan in attempt to calculate $\theta$ ✓<br>Doubles 12.4 / $\theta$ to get $24.8^\circ$ ✓ | Do not allow use of double slit equation.<br>In MP2 there must be some indication where the factor of 2 comes from. | 2    | 1 × AO1<br>1 × AO2 |

| Question | Answers                                                                                                                                                                                                      | Additional comments/Guidelines                                                                                                                                                            | Mark | AO      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------|
| 09.2     | Use of $d \sin(\theta) = n\lambda$ ✓<br>( $d \sin 12.4 = 2 \times 540 \times 10^{-9}$<br>$5.027 \times 10^{-6} \text{ m}$ )<br>Finding $\frac{1}{\text{their value of } d}$ ✓<br>Answer that rounds to 200 ✓ | accept substitution into formula or rearrangement; allow one error in substitution<br><br>Condone PoT error in MP2<br>Expect 199, condone $200 / 2.0 \times 10^2$ but not $2 \times 10^2$ | 3    | 3 × AO1 |

| Question | Answers                                                                                                                                                                                                         | Additional comments/Guidelines                                                                                                        | Mark | AO      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------|---------|
| 09.3     | Less diffraction or smaller $\theta$ and reference to relationship between angle and wavelength ✓<br>the new light source has a shorter wavelength ✓<br>Idea that colour changes towards blue end of spectrum ✓ | If no other mark awarded allow for one mark:<br>spots are closer<br>OR<br>distance between maxima decreased<br>OR<br>different colour | 3    | 3 × AO3 |
| Total    |                                                                                                                                                                                                                 |                                                                                                                                       | 8    |         |

| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

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|----|----------|----------------------------------------------------------|-----|
| 15 | <b>D</b> | increases the critical angle at the surface of the core. | AO1 |
|----|----------|----------------------------------------------------------|-----|

| Question | Answers                                                                                                                              | Additional comments/Guidelines           | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|-----|
| 07.1     | idea that oscillations occur parallel (to energy transfer) ✓<br>correct reference to oscillations and direction of energy transfer ✓ | Condone “vibrations” for “oscillations”. | 2    | AO1 |

| Question | Answers                                                                                                                                                                                                       | Additional comments/Guidelines                                                                            | Mark | AO                            |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------|-------------------------------|
| 07.2     | uses $v = \frac{x}{t}$ with a relevant time in seconds ✓<br><br>takes into account that their $\Delta t$ is for multiples of $2d$ ✓<br><br>answer in range $5.8 \times 10^{-3}$ to $6.1 \times 10^{-3}$ (m) ✓ | Relevant time should be between $2.2 \times 10^{-6}$ s and $2.6 \times 10^{-6}$ s, or multiples of these. | 3    | 1 × AO4<br>1 × AO3<br>1 × AO2 |

| Question | Answers                                                                                                                                                   | Additional comments/Guidelines                      | Mark | AO  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------|-----|
| 07.3     | time for ultrasound to travel through paint will be longer (than for equivalent thickness of metal) ✓<br><br>so will appear as a thicker piece of metal ✓ | If no other mark accept there is an extra distance. | 2    | AO3 |

| Question | Answers                                                                                                                                                         | Additional comments/Guidelines                                                 | Mark | AO  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------|-----|
| 07.4     | distance for reflection <b>R</b> is $2(y + d)$ <b>OR</b> distance for reflection <b>S</b> is $2(y + d) + 2y$ or $(4y + 2d)$ ✓<br><br>shows difference is $2y$ ✓ | MP1: Expressions must be accompanied by reference to path <b>R</b> or <b>S</b> | 2    | AO3 |
| Total    |                                                                                                                                                                 |                                                                                | 9    |     |



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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
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| 14 | B | 480 nm |  |
|----|---|--------|--|

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|----|---|-----|-----|--|
| 17 | D | 2.4 | 2.2 |  |
|    |   |     |     |  |

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|----|---|-------------|-------------|--|--|
| 22 | B | $n_x > n_y$ | $n_y < n_z$ |  |  |
|----|---|-------------|-------------|--|--|

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| 23 | D | The light from the grating must be polarised. |  |
|----|---|-----------------------------------------------|--|

| Question | Answers                                                                                                                                                                                                                              | Additional comments/Guidelines                                                                                                                                                                                                   | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 08.1     | <p>Spreading of pulse / parts of a pulse take different times to travel through the fibre / pulse broadening ✓</p> <p>Due to different paths through the optical fibre / due to entering the optical fibre at different angles ✓</p> | <p>MP1 must refer to pulse/signal/information etc. Eg 'different rays take different times' is insufficient for MP1</p> <p>Zero marks for a description that includes material dispersion or references to different colours</p> | 2    | AO1 |

| Question | Answers                                    | Additional comments/Guidelines | Mark | AO  |
|----------|--------------------------------------------|--------------------------------|------|-----|
| 08.2     | use of $v = \frac{c}{n}$ leading to 1.52 ✓ | Condone 2sf here               | 1    | AO1 |

| Question | Answers                                                                                                                                                                                             | Additional comments/Guidelines                                                                                                                                                                                | Mark | AO  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 08.3     | <p>Use of <math>\Delta x = v\Delta t</math> to find the maximum path difference ✓</p> <p>Path difference evaluated to at least 3sf, then adds 10 km to find the longest path to get 12.7 (km) ✓</p> | <p>MP1 requires words or algebra or subject of equation</p> <p>Alternative:<br/>Calculates total time ✓<br/>Evaluates total time and then calculates distance of longest path ✓</p> <p>MP2 depends on MP1</p> | 2    | AO2 |

| Question    | Answers                                                                                                     | Additional comments/Guidelines | Mark | AO  |
|-------------|-------------------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| <b>08.4</b> | (Use of $\cos^{-1} \left\{ \frac{10.0}{12.7} \right\}$ leading to) $38^\circ$ ✓<br><br>ecf from <b>08.3</b> |                                | 1    | AO2 |

| Question    | Answers                                                                                                                                                                                                                                                                                          | Additional comments/Guidelines | Mark | AO  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| <b>08.5</b> | Correct read-off for the time period ( $0.8 \times 10^{-4}$ s) (reject POT error here)<br><br><b>OR</b><br><br>use of $f = \frac{1}{T}$ (condone a read-off for an incorrect time period, eg pulse duration, interval between pulses) ✓<br><br>Both correct leading to $1.25 \times 10^4$ (Hz) ✓ |                                | 2    | AO3 |

| Question | Answers                                                                                                                                                                        | Additional comments/Guidelines | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| 08.6     | Read off pulse duration = $0.24 \times 10^{-4}$ s (reject POT error)<br>OR<br>Add the time delay ( $13.7 \times 10^{-6}$ s) to a value of time ✓<br>$3.8 \times 10^{-5}$ (s) ✓ |                                | 2    | AO3 |

| Question | Answers                                | Additional comments/Guidelines | Mark | AO  |
|----------|----------------------------------------|--------------------------------|------|-----|
| 08.7     | Idea of broadened pulses overlapping ✓ |                                | 1    | AO2 |
| Total    |                                        |                                | 11   |     |



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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
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| Question | Answers                                                         | Additional comments/Guidelines                                                                                                                             | Mark | AO  |
|----------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 09.1     | Angle is defined as $\frac{\text{arc length}}{\text{radius}}$ ✓ | Alternative:<br>( $\sin \theta$ or $\tan \theta = n \frac{w}{D}$ and)<br><br>angle is small so $\sin \theta$ (or $\tan \theta$ ) is approximately $\theta$ | 1    | AO3 |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                | Additional comments/Guidelines                                                                                                                                                                                                                                                                                                                      | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 09.2     | <p>Direct measurement of separation of at least 5 (angular) fringes with evidence seen on figure ✓</p> <p>Use of measurement of scale line to determine (angular) fringe width(s) ✓</p> <p>Use of <math>\theta_n = \frac{n\lambda}{s}</math> ✓</p> <p><math>1.05 \times 10^{-11}</math> to <math>1.25 \times 10^{-11}</math> (m) (at least 2 sf) ✓</p> | <p>MP1 only awarded when annotation shows measurements taken to centre of fringe.</p> <p>Expect angular fringe separation to be around <math>2.2 \times 10^{-5}</math> rad</p> <p>Do not allow diffraction grating equation</p> <p>MP3 is rejected if candidate attempts to evaluate a length for either w or D</p> <p>MP4 reliant on MP2 AND 3</p> | 4    | AO3 |

| Question     | Answers                                                                                                                           | Additional comments/Guidelines                                                                                                                      | Mark     | AO  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| <b>09.3</b>  | Use of $mv = \frac{h}{\lambda}$ (to find $v$ ) ✓<br>Use of $\frac{1}{2}mv^2 = eV$ ✓<br>$1.5 \times 10^4$ (V) ecf on <b>09.2</b> ✓ | MP1 and MP2 can be combined<br>Expect to see $V = \frac{h^2}{2\lambda^2 me}$<br>Expect $1.2\text{-}1.3 \times 10^4$ (V) if prev answer was in range | 3        | AO2 |
| <b>Total</b> |                                                                                                                                   |                                                                                                                                                     | <b>8</b> |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
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| Question | Answers                          | Additional comments/Guidelines                          | Mark | AO  |
|----------|----------------------------------|---------------------------------------------------------|------|-----|
| 11.1     | $x = 52.3 \text{ cm} \checkmark$ | Accept equivalent answer in m or mm<br>Must be 3sf only | 1    | AO1 |

| Question | Answers                                                                                                      | Additional comments/Guidelines        | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------|---------------------------------------|------|-----|
| 11.2     | 1 mm uncertainty leading to 0.2(%)<br>$\frac{0.1 \text{ (cm)}}{\text{answer to 11.1}} \times 100 \checkmark$ | Condone 0.19(%)<br><br>1 or 2 sf only | 1    | AO2 |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                      | Additional comments/Guidelines                                                                                                                                                                                                                    | Mark | AO                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|
| 11.3     | Any two of: $\checkmark\checkmark$ <ul style="list-style-type: none"> <li>Use of <math>\tan\theta = \frac{x}{2D}</math> to determine <math>\theta</math></li> <li>Use of <math>d\sin\theta = n\lambda</math> with their <math>(\sin)\theta</math> and their <math>d</math></li> <li>Correct expression of <math>d</math> (reject POT error)</li> </ul> $\lambda = 648 \text{ nm} \checkmark$ | Ecf on their $x$<br>For bullet point 1 only, condone $\sin\theta$ in place of $\tan\theta$ only if the candidate comments on small angle approximation in words or algebraically<br>Expect $7.45^\circ$<br><br>$d = 1.0 \times 10^{-5} \text{ m}$ | 3    | AO2 $\times 2$<br>AO3 |

| Question     | Answers                                                                                                                                                                       | Additional comments/Guidelines       | Mark     | AO  |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------|-----|
| 11.4         | Idea that the $x$ /angle between bright maxima will increase ✓<br>Decreases percentage uncertainty in $x$ or $\theta$<br>AND decreases percentage uncertainty in wavelength ✓ | Ignore references to grating spacing | 2        | AO4 |
| <b>Total</b> |                                                                                                                                                                               |                                      | <b>7</b> |     |

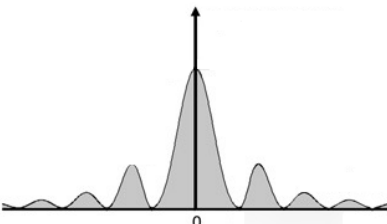
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| Question | Key | Answer | AO |
|----------|-----|--------|----|
|----------|-----|--------|----|

|    |   |     |     |
|----|---|-----|-----|
| 22 | C | 68° | AO1 |
|----|---|-----|-----|



|    |   |     |     |
|----|---|-----|-----|
| 23 | C | 28° | AO2 |
|    |   |     |     |

| Question | Answers                                                                                                                                                                                                           | Additional comments/Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark | AO  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 07.1     | <p>Basic description of pattern: central bright maximum with second maxima either side ✓</p> <p>Detail of central maximum e.g. brighter than secondary maxima; width is twice that of any secondary maximum ✓</p> | <p>Max 1 for marks awarded from an unlabelled diagram of visible pattern.</p> <p>A sketch of the visible pattern should have central maximum labelled.</p>  <p>A sketch of the intensity variation must have y-axis labelled as intensity</p>  <p>Reject any reference to white light or a continuous spectrum occurring.</p> | 2    | AO1 |

| Question | Answers                                                                                   | Additional comments/Guidelines                                                                                                                                      | Mark | AO  |
|----------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 07.2     | <p>Idea that pattern is broader ✓</p> <p>Brightness/intensity (of maxima) decreases ✓</p> | <p>MP1: accept a specific reference e.g. width of central maximum widens</p> <p>MP1: Reject reasoning based on double-slit interference or diffraction grating.</p> | 2    | AO1 |

| Question | Answers                                                                                                            | Additional comments/Guidelines                                           | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------|-----|
| 07.3     | Calculates $d$ ( $2.5 \times 10^{-6}$ m) <b>OR</b> substitutes values into diffraction equation ✓<br>Max $n = 4$ ✓ | Allow a POT in error in $d$ or $\lambda$ but not both.<br>( $n = 4.72$ ) | 2    | AO2 |
| Total    |                                                                                                                    |                                                                          | 6    |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers                                                                                                                                    | Additional comments/Guidelines        | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------|-----|
| 11.1     | Determines speed of light in glass as $1.97 \times 10^8 \text{ (m s}^{-1}\text{)}$ ✓<br><br>$1.03 \times 10^8 \text{ (m s}^{-1}\text{)}$ ✓ | MP1: Allow 2 sf value in working out. | 2    | AO2 |

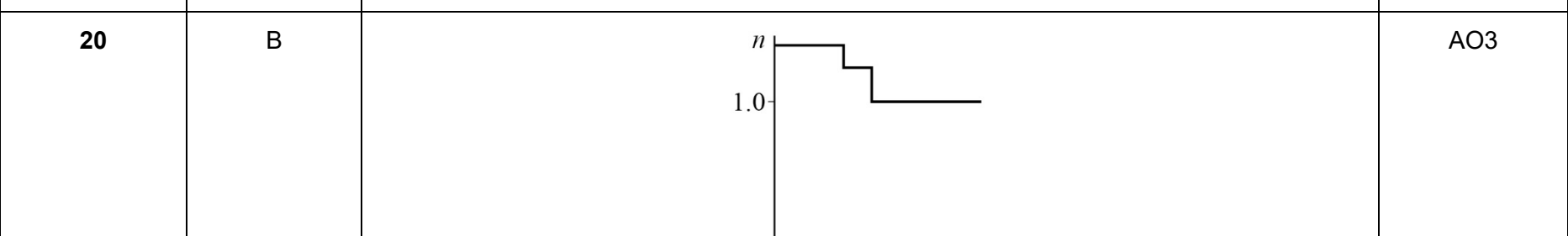
| Question | Answers                                                                                                                                    | Additional comments/Guidelines                                                                                                                              | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 11.2     | Angle of refraction = $61^\circ$ ✓<br><br>Uses Snell's law with either angle associated with the correct refractive index ✓<br><br>$1.2$ ✓ | MP2: e.g. $n_1 \sin \theta_1 = n_2 \sin 61$ <b>OR</b><br>$1.52 \sin 45 = n_2 \sin \theta_2$<br>Alternative MP2: Calculates $n_g = 0.809$<br><br>Expect 1.23 | 3    | AO2 |

| Question | Answers | Additional comments/Guidelines                          | Mark | AO  |
|----------|---------|---------------------------------------------------------|------|-----|
| 11.3     | $1.1$ ✓ | Must be correctly rounded.<br>Calculator value = 1.0748 | 1    | AO2 |

| Question     | Answers                                                                                                                                                                                                                                                                                                       | Additional comments/Guidelines                                                                                                                        | Mark     | AO  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| 11.4         | <p>If <math>n &lt; n_{\min}</math> the critical angle would be <math>&lt; 45^\circ</math> ✓</p> <p>Total internal reflection will occur (at first glass–material boundary) ✓</p> <p>Idea that light will not pass into the material or that the angle of deviation (from horizontal) cannot be measured ✓</p> | <p>MP1 may be a calculated example or a reference to critical angle equation.</p> <p>MP1: Allow “critical angle <math>&lt;</math> incident angle”</p> | 3        | AO4 |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                       | <b>9</b> |     |

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| Question | Key | Answer | AO |
|----------|-----|--------|----|
|----------|-----|--------|----|





| Question | Answers                                                                                                                                                                                                                                                          | Additional comments/Guidelines                                                                                                                                                | Mark | AO  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 02       | <p>Uses <math>d = \frac{10^{-3}}{280}</math> or equivalent ✓</p> <p>Uses <math>d \sin \theta = n \lambda</math> with <math>\theta = 90</math> <b>OR</b> <math>n \lambda = d</math> to find <math>n</math> ✓</p> <p>Doubles their <math>n</math> and adds 1 ✓</p> | <p>expect <math>3.6 \times 10^{-6}</math>; condone POT error</p> <p><math>n</math> must be integer for MP3</p> <p>(full marks for correct final answer = 11 bright spots)</p> | 3    | AO1 |
| Total    |                                                                                                                                                                                                                                                                  |                                                                                                                                                                               | 3    |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers                                                                                | Additional comments/Guidelines                                        | Mark | AO  |
|----------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|-----|
| 07.1     | Sketch shows the light totally internally reflecting multiple times in the water jet ✓ | Lines must be straight, judged by eye<br>Condone roughness of drawing | 1    | AO2 |

| Question | Answers                                                                                                            | Additional comments/Guidelines | Mark | AO  |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| 07.2     | Line travels straight to water–air boundary, then refracts downwards at boundary then travels straight to screen ✓ |                                | 1    | AO3 |

| Question | Answers                                                                                                                                                                                                                                                                                  | Additional comments/Guidelines | Mark | AO  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| 07.3     | <p>The angle of incidence of the light decreases and becomes less than the critical angle ✓</p> <p>so the light refracts out of the water / no longer totally internally reflects ✓</p> <p>Idea that the light speeds up so it bends away from the normal when it leaves the water ✓</p> |                                | 3    | AO3 |

| Question     | Answers                                                                                                                                                                                                                                                                                                                                                                                                          | Additional comments/Guidelines                                                                                                                  | Mark     | AO  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| <b>07.4</b>  | <p>Any <b>one</b> from: ✓</p> <ul style="list-style-type: none"> <li>• put a warning sign on the door</li> <li>• idea of not using lasers at eye level</li> <li>• do not point lasers at the audience <b>OR</b> do direct lasers at a screen</li> <li>• keep lasers stored safely (locked etc)</li> <li>• use lasers with an 'on' button that must be held down</li> <li>• use non-reflective screen.</li> </ul> | <p>Condone laser eye protection goggles</p> <p>Do not accept simply 'goggles' unless it is clear they are specifically for laser protection</p> | 1        | AO4 |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                 | <b>6</b> |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

|    |   |     |     |
|----|---|-----|-----|
| 20 | C | $y$ | AO1 |
|----|---|-----|-----|

|    |   |                                                        |     |
|----|---|--------------------------------------------------------|-----|
| 22 | c | $\frac{L}{v} \left( \frac{1}{\cos \theta} - 1 \right)$ | AO2 |
|----|---|--------------------------------------------------------|-----|

|    |   |                |     |
|----|---|----------------|-----|
| 23 | C | acrylic Teflon | AO1 |
|----|---|----------------|-----|



| Question | Answers                                                                                   | Additional comments/Guidelines        | Mark | AO  |
|----------|-------------------------------------------------------------------------------------------|---------------------------------------|------|-----|
| 01       | Correct expression for $d$ in m <b>OR</b> use of $n\lambda = d \sin \theta$ ✓<br>16 (°) ✓ | Condone incorrect $d$ in MP1 option 2 | 2    | AO1 |
| Total    |                                                                                           |                                       | 2    |     |

| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers | Additional comments/Guidelines | Mark | AO  |
|----------|---------|--------------------------------|------|-----|
| 04.1     | 1.32 ✓  | Condone 2sf                    | 1    | AO1 |

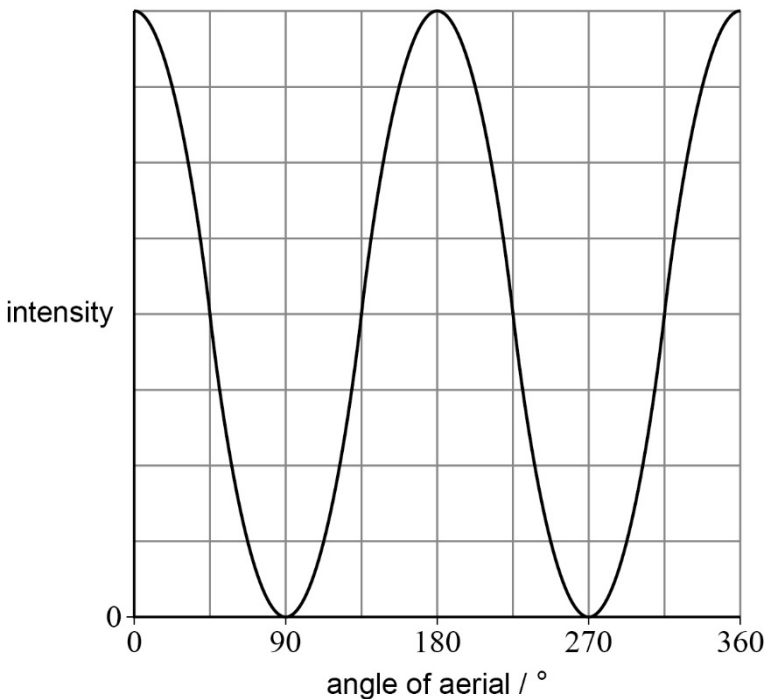
| Question     | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional comments/Guidelines                                                                                                                                           | Mark     | AO  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| 04.2         | <p>Any <b>two</b> from: ✓✓</p> <ul style="list-style-type: none"> <li>• more (directions of paths of) light refracted out of the fibre / fewer paths available / smaller range of angles / fewer lengths of path available / fewer (total internal) reflections</li> <li>• reduces <b>modal</b> dispersion</li> <li>• reduces pulse broadening</li> <li>• increases rate of data transmission / reduces (chance of) data loss</li> </ul> | <p>Ignore references to 'efficiency'</p> <p>Ignore references to speed of light down fibre</p> <p>Apply list rule</p> <p>Condone 'prevents' for 'reduces' throughout</p> | 2        | AO1 |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                          | <b>3</b> |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers                                                                                                                                              | Additional comments/Guidelines | Mark | AO                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------|--------------------|
| 08.1     | Use of $c = f\lambda$ to find $\lambda$<br><br><b>OR</b><br>Use of $w = \frac{\lambda \times 0.75}{0.095}$ with their wavelength ✓<br><br>0.22 (m) ✓ |                                | 2    | 1 × AO2<br>1 × AO3 |

| Question | Answers                                                                                                                                   | Additional comments/Guidelines             | Mark | AO  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------|-----|
| 08.2     | Idea that this equation requires the distance to the screen to be large (compared with $w$ or $s$ or $\lambda$ ) <b>OR</b> small angles ✓ | Ignore reference to the width of each slit | 1    | AO3 |

| Question | Answers                                                                            | Additional comments/Guidelines                                                                                                                                                                                                    | Mark | AO  |
|----------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 08.3     | Sketch: two peaks and two troughs of zero intensity per 360 degrees (labelled A) ✓ | <p>Ignore shape apart from number of peaks.<br/>Ignore phase.<br/>Condone troughs that do not go to zero intensity</p> <p>Expect to see:</p>  | 1    | AO2 |

| Question | Answers                                                                                                                                                                                                          | Additional comments/Guidelines                                                                                                      | Mark | AO  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 08.4     | Idea that the aerial must be aligned with (the oscillations of) the wave ✓<br><br>Idea of (the aerial and the wave) being parallel leading to high intensity and being perpendicular leading to zero intensity ✓ | Condone 'vibrations' for oscillations<br><br>Condone references to the gap orientation for the wave oscillation orientation for MP2 | 2    | AO2 |

| Question     | Answers                                                                                    | Additional comments/Guidelines | Mark     | AO  |
|--------------|--------------------------------------------------------------------------------------------|--------------------------------|----------|-----|
| 08.5         | Flat line (labelled B) ✓<br><br>at approximately half height of line <b>A</b> 's maximum ✓ |                                | 2        | AO3 |
| <b>Total</b> |                                                                                            |                                | <b>8</b> |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

|    |   |                                                                 |     |
|----|---|-----------------------------------------------------------------|-----|
| 19 | B | directing the lasers so that patterns are produced at eye level | AO4 |
|----|---|-----------------------------------------------------------------|-----|



|    |   |               |     |
|----|---|---------------|-----|
| 21 | A | $\varphi$ $g$ | AO3 |
|----|---|---------------|-----|

Unknown Unknown - Unit 2 - Q01

| Question | Marking guidance                                                     | Mark   | Comments |
|----------|----------------------------------------------------------------------|--------|----------|
| 01.1     | <u>cladding</u>                                                      | 1      |          |
| 01.2     | $\sin \theta_c = 1.41/1.46$ ✓<br>$\theta_c = 75$ (degrees) (74.96) ✓ | 1<br>1 |          |
| 01.3     | 65 (degrees)                                                         | 1      |          |

Unknown Unknown - Unit 2 - Q01 (continued 2)

|                                                                                                  |   |
|--------------------------------------------------------------------------------------------------|---|
| $R = \frac{V}{I}$ $= \frac{1.7 \times 10^{-8} \times 1.4}{7.8 \times 10^{-7}}$ $= 0.031(\Omega)$ | 1 |
|                                                                                                  | 1 |